MTH 252Z-401: Integral Calculus Oregon State University

Summer 2026

Instructor: Michael Gilliam (Dr G)
Lecture: Recordings on Canvas
Office: Kidd 256
Prerequisite: MTH 251Z, MTH 251HZ, MTH 251, or MTH 251H
Email: gilliamm@oregonstate.edu
Credits: 4 (offered F, W, S & U quarters)
Office Hours: Zoom, by appointment
CRN: 74136

CATALOG DESCRIPTION: Explores Riemann sums, definite integrals, and indefinite integrals for real-valued functions of a single variable. Explores topics graphically, numerically, and symbolically in real-life applications. Emphasizes abstraction, problem-solving, modeling, reasoning, communication, connections with other disciplines, and the appropriate use of technology.

MTH 252Z STUDENT LEARNING OUTCOMES: A successful student in MTH 252Z will be able to:

1. Approximate definite integrals using Riemann sums and apply this to the concept of accumulation and the definition of the definite integral.
2. Explain and use both parts of the Fundamental Theorem of Calculus.
3. Choose and apply integration techniques including substitution, integration by parts, basic partial fraction decomposition, and numerical techniques to integrate combinations of power, polynomial, rational, exponential, logarithmic, trigonometric, and inverse trigonometric functions.
4. Use the integral to model and solve problems in mathematics involving area, volume, net change, average value, and improper integration.
5. Apply integration techniques to solve a variety of problems, such as work, force, center of mass, or probability.

REQUIRED COURSE CONTENT:
At the end of the course, students will be able to…

1. Approximate definite integrals using Riemann sums and apply this to the concept of accumulation and the definition of the definite integral.
    a. Students will be able to express finite sums using sigma notation.
    b. Students will be able to use Riemann sums to describe the process of approximating the net signed area between a curve and an axis.
2. Explain and use both parts of the Fundamental Theorem of Calculus.
    a. Students will be able to recognize and express the definite integral as a limit of a Riemann sum.
    b. Students will use and compare different methods for calculating definite integrals, such as linear properties of integrals, net-signed area, and graphical approaches.
    c. Students will explain and apply the concept of indefinite integrals and its connection to antidifferentiation.
    d. Students will explain the connection between derivatives and integrals and apply their understanding using the Fundamental Theorem of Calculus.

3. Choose and apply integration techniques including substitution, integration by parts, basic partial fraction decomposition, and numerical techniques to integrate combinations of power, polynomial, rational, exponential, logarithmic, trigonometric, and inverse trigonometric functions.
    a. Students will be able to integrate power, polynomial, rational, exponential, logarithmic, trigonometric, and inverse trigonometric functions using basic rules.
    b. Students will be able to use substitution and integration by parts to algebraically integrate appropriate combinations of functions.
    c. Students will be able to use partial fraction decomposition to evaluate integrals of rational functions whose denominators may be expressed as products of distinct linear factors.
    d. Students will be able to use numerical techniques, such as Midpoint, Trapezoidal, and Simpson's rules, to approximate definite integrals.
4. Use the integral to model and solve problems in mathematics involving area, volume, net change, average value, and improper integration.
    a. Students will be able to use definite integrals to find the area between two curves.
    b. Students will be able to calculate volumes of solids, such as solids of revolution or prisms, using integrals.
    c. Students will be able to apply the integral to find the average value of a function over an interval.
    d. Students will be able to apply the integral to find the net change of a function over an interval.
    e. Students will be able to recognize, describe, and calculate improper integrals.
5. Apply integration techniques to solve a variety of problems, such as work, force, center of mass, or probability.

    Students will apply integration to problems in the instructor's choice of context, including but not limited to the possible options above. At least two distinct applications are recommended based on the population of students in the class.

COURSE CONTENT:

Week 1: Riemann sums
Week 2: Definite integrals
Week 3: Definite integrals and Antiderivatives
Week 4: Fundamental Theorem of Calculus
Week 5: Techniques of Integration: Substitution and Exponentials & Logs
Week 6: Techniques of Integration: Integration by Parts & Trig Functions
Week 7: Techniques of Integration: Trig Substitution & Partial Fractions
Week 8: Applications: Areas between Curves & Volumes by Slicing
Week 9: Applications: Cylindrical Shells, Arc Length & Physics
Week 10: Applications: Physics and Improper Integrals

Diversity/Inclusion Statement (Personal):
It is my intent to make an equitable and inclusive course for all students, where all feel welcome, respected and appreciated. The diversity of students, in terms background, perspective and lived experience is an immense resource, as we can learn and grow by constructively communicating and collaborating with each other. It is always my intent to present materials and activities that are respectful of race, ethnicity, gender identity, sexual orientation, disability, age, socioeconomic factors, religion, culture, perspective, and other background characteristics. Your suggestions about how to improve the equity, justice, and inclusiveness of this course are encouraged and greatly appreciated.

LEARNING RESOURCES

Required Learning Resources:

Textbook:	OpenStax College. (2016). Calculus, Volume 2. OpenStax. https://openstax.org/details/books/calculus-volume-2?Book%20details (CC-BY-NC-SA) Free online pdf
Achieve:	See our Canvas page for Achieve registration instructions. Access to Achieve can be purchased through the Beaver Store link in Canvas or directly from the Beaver Store.
Internet-Capable Device:	We will be using CANVAS for much of our communication outside of class time. Achieve is our required online homework platform, as well as Gradescope, and other internet applications

AI Course Policy: Use of generative AI (genAI) tools such as ChatGPT is permitted in this course for specific uses. Continue reading through this policy for a list. You should be able to reproduce any submitted work without genAI.

Violation of this policy is a breach in Academic Misconduct and is cheating.

Permitted uses of genAI:
- Generate practice problems when studying
- Asking for an alternate explanation of a concept you did not understand in class
- Asking for feedback on an explanation you wrote
- Looking up definitions (e.g., [the moment of inertia for an object, trig identities])

Not prohibited, but not recommended uses of genAI:
- Generate study materials, including note sheets, concept maps, generalized study guides, etc.
- Making an equation sheet
- Asking for a solution to a homework question to check your work

Prohibited uses of genAI:
- Asking for a solution to a homework question and submitting it for a grade
- Asking for an answer to an in-class clicker question
- Asking for a solution to an exam question
- Asking for an answer / solution to any course assignment type that you have not first attempted

Recommended Learning Resources:

Riemann Sum Calc:	Required (free) access to the class Riemann Sum calculator is needed. (Not needed on exams.) Link: https://www.geogebra.org/m/ezspy4zu
DESMOS Account:	*This is recommended. Desmos is an online graphing tool which we will often use in class. You can visit Desmos.com and there is a free smartphone app, as well.*
MSLC:	A virtual Math & Stats Learning Center will be open for tutoring beginning Week 2!

Evaluation of Student Performance: Your grade and measurement of your progress on the course outcomes will be based various course activities, recitation "participation," weekly online homework, bi-weekly written homework, and bi-weekly exams.

Assessment Type	Percent	Notes
Participation	**7%**	
--Discussion Reflection --Reading Quizzes	4% 3%	Participating in each Discussion Board assignment Reading quizzes covering content from OpenStax V2
Homework	**40%**	
--Achieve-Online HW --WHW	20% 20%	The online homework system Achieve Written homework assignments every week
Mini-EXAMS	**53%**	
--Mini-Exams 1, 2, 3, 4 --Mini-Exam 5	10% each 13%	Weeks 3, 5, 7, 9 – Tuesday-Thursday, Remote, 75 min. timed exams Week 11 – Monday - Wednesday, Remote, 110 min. timed exam

Grading Scale.

Letter Grade	Percent Range
A	[100 - 92] %
A-	(92 - 90] %
B+	(90 - 88] %
B	(88 - 82] %
B-	(82 - 80] %
C+	(80 - 78] %
C	(78 - 72] %
C-	(72 - 70] %
D+	(70 - 67] %
D	(67 - 60] %
F	Below 60%

SECRETS TO SUCCESS:

1) ***Attitude** is vital in mathematics.*
   - Be positive! Keep trying!
   - Be respectful of everyone: classmates, GTA, instructor, tutors.
   - Support and resources are available to help you succeed.
   - You will do well if you have the right attitude.
2) *Engage in **Active-Learning***
   - Throughout the term, you will be engaged in solving problems as a part of the class structure.
   - Please come to class prepared with your workbook, calculator, and your attention.

ASSESSMENT INFORMATION

At the end of the term, final grades will be based on the above categories and weights, calculated and rounded to the nearest tenth place, and assigned a letter grade, according to the scale above. **Gradescope Submissions:** Students are required to verify their personal submissions in Gradescope by refreshing the page to see that all the pages have been uploaded. Since verifying submissions is so accessible, it is assumed that submissions made are exactly in the form the student intended. **Warning:** Gradescope sends a confirmation email EVEN IF the submission is not accepted, which is why students need to refresh to verify their own submissions.

1. Participation (7%)

- ***Discussion Reflection (DR) 4%***: We will have weekly discussion reflection (DR#) assignments. You will be required to participate in each DR by making one or more posts.
  - ***Drop Policy: One of the lowest scores*** *will be dropped.*
  - ***Late Policy****:* Late work is accepted with notice at or before the due date. Flexibility will be considered for extenuating circumstances.
- **Pre-Class Reading Quizzes (RQ) 3%**: We will have several pre-class reading quizzes (RQ#). These quizzes are short, require no previous knowledge of the material, require no calculations nor algebra, and come directly from reading/skimming the corresponding section in the OpenStax course textbook. To support success, each student receives **two attempts** per quiz. Each attempt is a full retake.
  - **Drop Policy**: **One of the lowest scores** will be dropped.
  - ***Late Policy****:* Notice is to be given before or on the due date. RQ extensions will be granted for extenuating circumstances. Flexibility will be considered for extenuating circumstances.

2. Homework (40%)

- ***Achieve (A) 20%:*** This is an on-line homework system. Achieve is set up for you to practice the skills you will need to develop as we proceed through each section covered in MTH252.
  - ***Purchases:*** Single-Term Achieve access code. Purchases can be made either through the OSU Beaver Store, by directly by clicking on an assignment in Canvas. See you course canvas page for details.
  - ***Drop Policy:*** **Two of the lowest** scores will be dropped.
  - **Late Policy:** Late work is accepted with the penalty of 3% per day late, applying only to problems completed after the due date. All assignments will remain open for 34 days after the due date or until midnight the day of our final, whichever comes first**.**
- ***Written Homework (WHW) 20%:*** These are weekly written assignments, designed to strengthen your ability to write full solutions to problems. Being able to articulate oneself through written proofs is essential in STEM fields. **One of the lowest** scores will be dropped.
  - ***Drop Policy:*** **One of the lowest** scores will be dropped.
  - ***Submissions:*** **Submit whatever you have done by the due date**. You can resubmit further if given an extension.
  - ***Late Policy:*** Late work is accepted with notice at or before the due date because solutions are posted on the due date/time to promote learning. Flexibility will be considered for extenuating circumstances.
  - ***Late Submissions:*** Late submissions without prior notice are accepted **within two weeks of the original due date**. As solutions will be posted near the due date, the penalty for no notice before or at the time of the due date is 50%.

3. **<u>Mini-Exams</u>** (55%) Online/Remote (and self-proctored), available between Tuesday 12am to Thursday 11:59pm and are timed.
   - ***<u>Mini-Exams 1 - 4</u>*** (10% each, total 40%) Weeks: 3, 5, 7, 9.
     These 75-minute timed mini-exams will occur every other week. These are available to complete anytime between Tuesday 12am and Thursday 11:59pm (PST). Exams will cover specific learning objectives from each corresponding Unit, shared below, and generally needed pre-req material.
     - **Mini-Exam Accommodations**: To support student success, we invite students with special circumstances to request accommodations/extensions by providing 48-hour notice through OSU email.
     - **Mini-Exam Retakes**: To support student learning, time will be given <u>during finals week</u> to retake (or make up) one of the mini-exams from 1 – 4. The score on the retake will fully replace the original exam score **ONLY IF it improves** their **overall grade**. A retake during finals will NOT lower a student's grade.
       - **Retake Date**: Available between Tuesday 12am to Thursday 11:59pm.
     - **Mini-Exam Make ups:** Are to be coordinated with the instructor.

   - ***<u>Mini-Exam 5</u>*** *(13%) partially cumulative.*
     This exam will cover Unit 5 material from weeks 9 and 10 along with select review material from prior mini-exams.

<u>Mini-Exam Policies.</u>

- Please request an alternative exam time from your instructor at least 1 week in advance in the case of conflict with a mini-exam or final exam time.
- Students may NOT use a calculator nor the textbook or any notes during the mini-exams. These are intentionally designed so calculators are not needed, with exact (not rounded) answers being emphasized.
- The mini-exam 5 will be given according to OSU guidelines in Week 11.
- Downloads and uploads for the mini-exams will done using the online platform Gradescope.
- In this remote setting, **online submissions require careful formatting**. Students are required to collaborate with the instructor to get access to, learn, and attentively use any scanning software to upload content using the appropriate formatting instructions.
- **ALL submissions** will be made as **<u>ONE pdf with multiple pages</u>**.
- **Submission issues?** Email your instructor <u>immediately AND attach</u> your exam as a pdf (or any other format) to that email.
- **Late mini-exam submissions** Submissions must be made through an email to the instructor.
  a. Exams submitted on time – no late penalty.
  b. Exams submitted up to 5 minutes late – 7.5% penalty is given.
  c. Exams submitted between 5 and 30 minutes late – 30% penalty is given.
  d. Exams will not be accepted more that 30 minutes late.
- Regrade requests will be available in Gradescope for a specified period after each exam.

How to Study for this Course (Ideas): (Optional to keep or move to Canvas)

- Attend lectures regularly, ideally bring the lecture notes printed or on a tablet. This gives you more time in class to think and practice while spending less time writing.
- Study/review your lecture notes and read through examples from the corresponding textbook section. Identify very specific things you do not understand. Get help to fill in these details asap.
  a. For each concept or problem type, clearly write out or identify a 3-to-4 step problem-solving process. Think general, use vocabulary, avoid focusing on specific numbers.

- Start homework after studying your notes and try to apply the "recipe" problem-solving process from above.
- Attend office hours, SI tables, or MSLC to get your questions answered. Review as needed.
- Continue completing homework, while progressively using less (and less) of your notes and peer support.
- Complete many problems during the week WITHOUT using any notes, without peer support, and without external support at all. This is testing your true knowledge, in preparation for the mini-exam.

Course Policies

- ***Canvas***.
  Canvas is used to communicate with students. Important class materials, including assignments, points earned, and announcements of any changes to the class schedule are made on Canvas. You are expected to check Canvas regularly and are responsible to be aware of any announcements on Canvas.

- ***E-Mail***.
  Email is the best and quickest way to reach me if you have a question or need to schedule an appointment. I check my ONID email frequently. You MUST begin any email pertaining to the class with "252" in the subject line. It is also best to email me at the address give at the beginning of this syllabus. I will try to respond within 24 hours. Due to privacy concerns, I only read and respond to emails sent from your OSU ONID email account, and grades will not be discussed via email. **Students are held accountable for checking their OSU email regularly**.

- ***Policy on Incompletes***.
  An incomplete grade is given only in accordance with the University grade policy.
  Link: https://registrar.oregonstate.edu/incomplete-grade-policy

OREGON STATE UNIVERSITY POLICY STATEMENTS

Academic Calendar.
All students are subject to the registration and refund deadlines as stated in the Academic Calendar: https://registrar.oregonstate.edu/osu-academic-calendar

Statement Regarding Religious Accommodation.
Oregon State University is required to provide reasonable accommodations for employee and student sincerely held religious beliefs. It is incumbent on the student making the request to make the faculty member aware of the request as soon as possible prior to the need for the accommodation. Link: Religious Accommodation Process for Students

Statement Regarding Students with Disabilities.
Accommodations for students with disabilities are determined and approved by Disability Access Services (DAS). If you, as a student, believe you are eligible for accommodations but have not obtained approval please contact DAS immediately at 541-737-4098 or at the link above. DAS notifies students and faculty members of approved academic accommodations and coordinates implementation of those accommodations. While not required, students and faculty members are encouraged to discuss details of the implementation of individual accommodations. Link: http://ds.oregonstate.edu

Student Conduct Expectations.

Reach Out for Success: University students encounter setbacks from time to time. If you encounter difficulties and need assistance, it's important to reach out. Consider discussing the situation with an instructor or academic advisor. Learn about resources that assist with wellness and academic success at oregonstate.edu/ReachOut. If you are in immediate crisis, please contact the Crisis Text Line by texting OREGON to 741-741 or call the National Suicide Prevention Lifeline: 1-800-273-TALK (8255) Link:

http://studentlife.oregonstate.edu/code

Policy on Incompletes.

Incompletes are given only in accordance with the University incomplete grade policy.

Reach OUT for Success.

University students encounter setbacks from time to time. If you encounter difficulties and need assistance, it's important to reach out. Consider discussing the situation with an instructor or academic advisor. Learn about resources that assist with wellness and academic success. Ecampus students are always encouraged to discuss issues that impact your academic success with the Ecampus Success Team. Email: ecampus.success@oregonstate.edu to identify strategies and resources that can support you in your educational goals. If you feel comfortable sharing how a hardship may impact your performance in this course, **please reach out to me as your instructor**.

Mental Health.

Learn about counseling and psychological resources for Ecampus students. If you are in immediate crisis, please contact the Crisis Text Line by texting OREGON to 741-741 or call the National Suicide Prevention Lifeline at 1-800-273- TALK (8255).

Financial Hardships. Any student whose academic performance is impacted due to financial stress or the inability to afford groceries, housing, and other necessities for any reason is urged to contact the Director of Care for support (541-737-8748).

Academic Misconduct.

Any action that misrepresents a student or group's work, knowledge, or achievement, provides a potential or actual inequitable advantage, or compromises the integrity of the educational process. Prohibited behaviors include, but are not limited to doing or attempting the following actions:

Link: OSU's Student Conduct Code

a. **Cheating**. Unauthorized assistance, or access to or use of unauthorized materials, information, tools, or study aids. Examples include, but are not limited to, unauthorized collaboration or copying on a test or assignment, using prohibited materials and texts, unapproved use of cell phones, internet, or other electronic devices, etc.

b. **Plagiarism**. Representing the words or ideas of another person or presenting someone else's words, data, expressed ideas, or artistry as one's own. Examples include, but are not limited to, presenting someone else's opinions and theories as one's own, using another person's work or words (including unpublished material) without appropriate source documentation or citation, working jointly on a project and then submitting it as one's own, etc.

c. **Falsification. Fabrication or invention of any information**. Examples include, but are not limited to, falsifying research, inventing or falsely altering data, citing fictitious references, falsely recording or reporting attendance, hours, or engagement in activities such as internships, externships, field experiences, clinical activities, etc.

d. **Assisting**. Any action that helps another engage in academic misconduct. Examples include, but are not limited to, providing materials or assistance without approval, altering someone's work, grades or academic records, taking a test/doing an assignment for someone else, compelling acquisition, selling, bribing, paying or accepting payment for academic work or assistance that contributes to academic misconduct, etc.

e. **Tampering**. Interfering with an instructor's evaluation of work by altering materials or documents, tampering with evaluation tools, or other means of interfering.

f. **Multiple submissions of work**. Using or submitting work completed for another or previous class or requirement, without appropriate disclosure, citation, and instructor approval.

g. **Unauthorized recording and use**. Recording and/or dissemination of instructional content without the express permission of the instructor(s), or an approved accommodation coordinated via Disability Access Services.

We will be moving through the Textbook in approximately the following order:

Unit 1

1.1 – Approximating Area / 3.6 – Numerical Integration
1.2 – The Definite Integrals

Unit 2

1.3 – The Fundamental Theorem of Calculus (part 1, 2) /4.10 – Antiderivatives (in Volume 1)
1.4 – Integration Formulas and the Net Change Theorem
1.5 – Substitution
1.6 – Integrals Involving Exponential and Logarithmic Functions

Unit 3

3.1 – Integration by Parts
3.2 – Trigonometric Integrals
3.3 – Trigonometric Substitution
3.4 – Partial Fractions

Unit 4

2.1 – Area between Curves
2.2 – Determining Volume by Slicing
2.3 – Volumes of Revolution: Cylindrical Shells
2.4 – Arc Length of a Curve (Not covering surface area)

Unit 5

2.5 – Physical Applications
3.7 – Improper Integrals